

Siddharth Korukonda

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EDUCATION

Stony Brook University

2023 – 2027

Bachelor of Science in Computer Science, Bachelor of Arts in Economics (Dean's List)

Stony Brook, NY

Minor in Applied Mathematics and Statistics, Specialization track in Artificial Intelligence and Machine Learning

Courses: Machine Learning, Artificial Intelligence, NLP, Distributed Systems, Computer Vision, Software Security, Cloud Computing, Advanced OS

AWARDS AND CERTIFICATIONS

Mathematical Association of America (MAA): AIME - 5 (AMC 12B - 106.5)

Columbia DivHacks 2025: Best Hack - Entertainment, Best Usage of Opik

SBUHacks 2025: Best Usage of NeuralSeek, Best Usage of Snowflake

JP Morgan Chase: Quantitative Reserach Job Simulation, Software Engineer Job Simulation

Amazon: Certified Cloud Practitioner (in progress)

Google: Data Analytics Professional (in progress)

TECHNICAL SKILLS

Languages: Java, Kotlin, Python, MySQL, C/C++, C#, Lua, JavaScript, HTML/CSS, Bash, TypeScript, LaTeX, R, MIPS

Frameworks: React.js, Node.js, Express.js, .NET, Django, Spring Boot, MongoDB, PyTorch, Flask, Tailwind

Developer Tools: Google Colab, Postman, Git, LangChain, PyCharm, IntelliJ, VS Code, Kubernetes, Docker, AWS CLI, Opik, Snowflake

Libraries: NumPy, Matplotlib, Scikit, SciPy, OpenCV, TensorFlow, OpenPose, VADER, ONNX Runtime, Transformers, Pandas

Leadership: Computer Vision Mentor SBU Robotics, SBU ICPC Mentor, Teaching Assistant System Fundamentals

EXPERIENCE

Software Developer

Jan 2025 – Present

Stony Brook University Hospital

Stony Brook, NY

- Design and build the hospital website in React and Flask, create reusable UI components, develop secure APIs, manage SQL and NoSQL data, implement login and roles, and keep pages fast and accessible.
- Collaborate with BS/MD and pre-med students to analyze de-identified patient records, curate datasets, and train/evaluate clinical AI models, incorporating feedback to improve model quality.
- Write clear docs, help teammates and staff use the tools, watch error logs, and fix issues to keep everything stable.

Secure Distributed Computation and Learning Networks Research

August 2024 – Present

Undergraduate Researcher

Stony Brook, NY

- Maintained 80% consensus in Byzantine networks by projecting updates onto convex sets and eliminating adversarial values
- Achieved 77% valid input preservation in real-time decision-making under adversarial corruption
- Enabled stable updates with 93% agent interaction through controlled neighborhood protocols

Undergraduate Teaching Assistant

August 2024 – May 2025

Stony Brook University

Stony Brook, NY

- Created assignments and labs for CSE 220 (System Fundamentals I).
- Delivered assistance to students in lectures of approximately 200 students.
- Graded assignments and held weekly recitations and office hours.

PROJECTS

SurveiLens - Real-Time AI Safety Platform

November 2025

React, Flask, NeuralSeek, Snowflake, ElevenLabs, OpenCV, Gemini, VADER, WebRTC/RTSP, Vultr

- Integrated OpenCV video analysis and ElevenLabs audio transcription through Gemini reasoning to generate real-time risk assessments
- Enhanced detection precision by combining object tracking, action recognition, and VADER sentiment to minimize false positives
- Enabled multi-device connectivity by streaming RTSP/WebRTC snapshots to Flask and syncing event data into Snowflake for analysis
- Leveraged NeuralSeek to coordinate multimodal AI pipelines, enforce risk-based policies, and broadcast alerts to the live monitoring dashboard

FastFacts - Real-Time AI Fact-Checking Desktop App

October 2025

Electron, React, Flask, Opik, LangChain, Tavily, Gemini, OpenAI Speech-to-Text

- Developed a desktop application that performs real-time fact-checking on live system audio using an agentic AI workflow.
- Implemented an automated pipeline with OpenAI Speech-to-Text for transcription, Gemini for claim detection, and Tavily MCP for retrieving reputable online sources.
- Used Gemini to analyze evidence and classify claims as true, false, or unsubstantiated, returning a final verdict upon session completion.
- Integrated Comet's Opik to evaluate LLM outputs, ensuring consistency, reliability, and explainability across all agent interactions.

SBU Care Connect - Website App for the Stony Brook Hospital Free Clinic

January 2025

React, Map API, Flask, SQL (partners & patients), NoSQL (chat logs), OpenAI API, Auth/JWT

- Built a React frontend with a map API that lets patients search the nearest Stony Brook Hospital partner by address, and added a chat box powered by an OpenAI LLM for patient guidance.
- Developed a Python Flask backend with an SQL database for partner locations and a separate SQL database for patient records, and used a NoSQL store for patient chat logs with APIs that serve map searches and chat sessions.
- Implemented authentication and login tied to a user key in SQL, secured data storage and access controls, and personalized features and content for each patient after sign in.